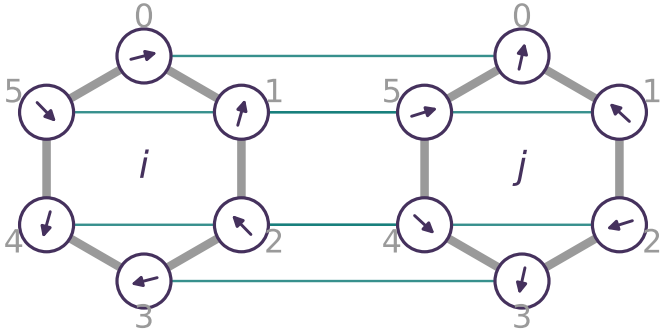
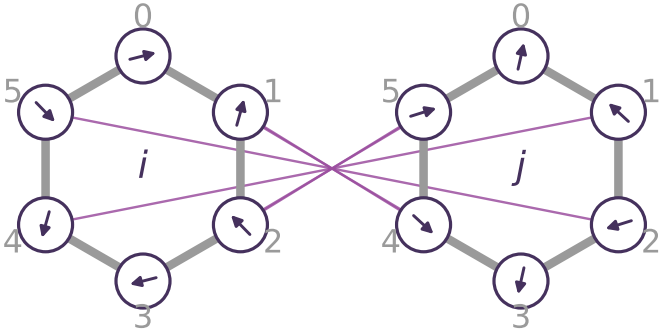
(a) positive edge: group a to a



(b) negative edge: group a to $a+3$



(c) cone point; nearby balanced state

3 at 0,
3 at π

closes up
head to tail